



**reversible reactions and
equilibrium**

Name: _____

Class: _____

Date: _____

Time: **68 minutes**

Marks: **64 marks**

Comments:

1.

This question is about a reversible reaction.

The reaction between solutions of iron(III) ions (Fe^{3+}) and thiocyanate ions (SCN^-) is reversible.

The ionic equation for the reaction is:



The colour of the equilibrium mixture is orange at room temperature.

(a) Give the name of the solvent used to dissolve the ions in this reaction.

(1)

(b) A few drops of a colourless solution containing a high concentration of thiocyanate ions (SCN^-) are added to the orange equilibrium mixture.

Explain the colour change observed.

(3)

(c) A water bath is set up at a temperature above room temperature.

When a test tube containing the orange equilibrium mixture is placed in the water bath, the mixture becomes more yellow.

Explain what this shows about the energy change for the forward reaction.

(3)

- (d) Explain why a change in pressure does **not** affect the colour of the equilibrium mixture.

(2)

- (e) Other metal ions form coloured equilibrium mixtures with thiocyanate ions.

Which metal ion could form a coloured equilibrium mixture with thiocyanate ions?

Tick (✓) **one** box.

Al^{3+}

☐

Co^{2+}

☐

Mg^{2+}

☐

Na^{+}

☐

(1)

(Total 10 marks)

2.

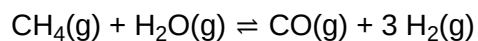
This question is about reversible reactions and equilibrium.

Hydrogen is used to produce ammonia in the Haber process.

The hydrogen is made in two stages.

Stage 1 is the reaction of methane and steam to produce carbon monoxide and hydrogen.

The equation for the reaction is:



- (a) Calculate the atom economy for the formation of hydrogen in **stage 1**.

Relative atomic masses (A_r): $\text{H} = 1$ $\text{C} = 12$ $\text{O} = 16$

Atom economy = _____ %

(2)

- (b) Explain why a low pressure is used in **stage 1**.

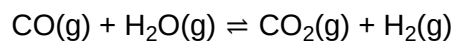
Give your answer in terms of equilibrium.

(2)

- (c) **Stage 2** uses the carbon monoxide produced in **stage 1**.

The carbon monoxide is reacted with more steam to produce carbon dioxide and more hydrogen.

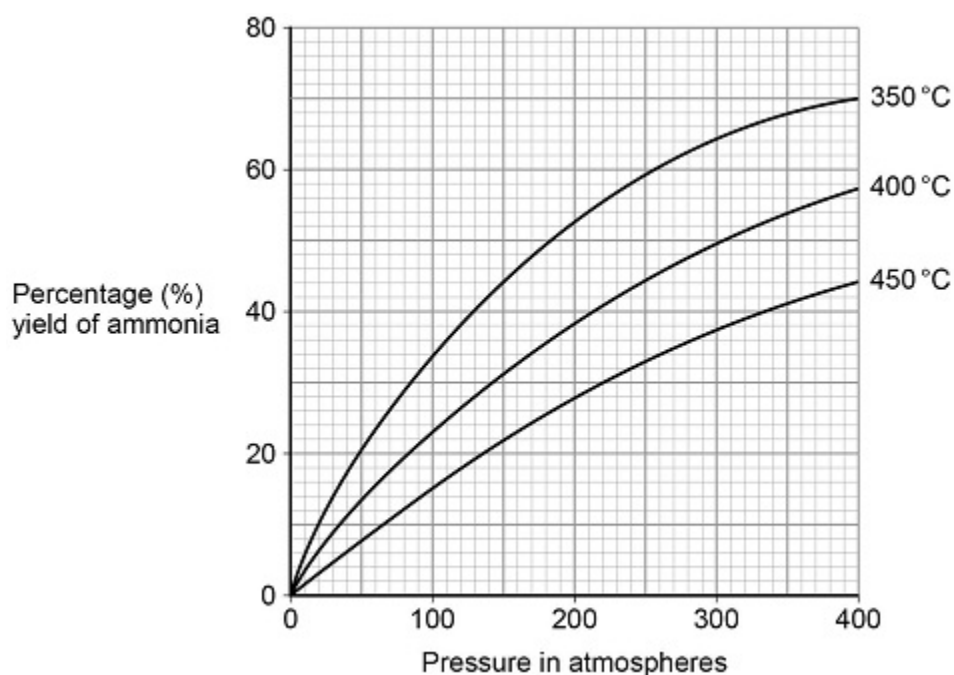
The equation for the reaction in **stage 2** is:



What is the effect of increasing the pressure on the equilibrium yield of hydrogen in **stage 2**?

(1)

The graph below shows the percentage yield of ammonia produced at different temperatures and pressures in the Haber process.



A temperature of 450 °C and a pressure of 200 atmospheres are used in the Haber process.

- (d) A student suggested that a temperature of 350 °C and a pressure of 285 atmospheres could be used instead of those used in the Haber process.

Determine how many times greater the percentage yield of ammonia obtained would be.

Use the graph.

Percentage yield = _____ times greater

(3)

- (e) A pressure of 285 atmospheres is **not** used in the Haber process instead of 200 atmospheres.

Give **one** reason why.

(1)

- (f) How does the graph above show that the forward reaction in the Haber process is exothermic?

(1)

- (g) World production of ammonia is now about 30 times greater than it was in 1950.

Suggest why the demand for ammonia has increased.

(2)

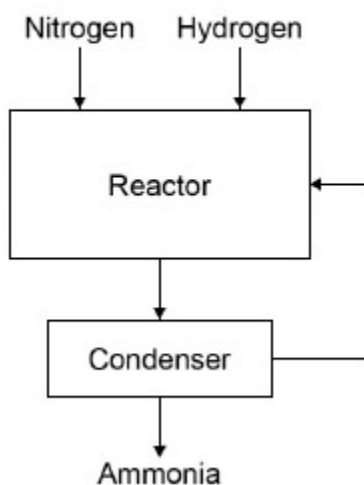
(Total 12 marks)

3.

Nitrogen and hydrogen react to produce ammonia in the Haber process.

Figure 1 shows the Haber process.

Figure 1



A gaseous mixture of ammonia, hydrogen and nitrogen leaves the reactor.

Table 1 shows the boiling points of the gases.

Table 1

Gas	Boiling point in °C
Ammonia	-33
Nitrogen	-196
Hydrogen	-253

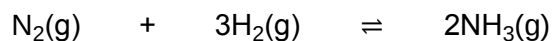
(a) Suggest how ammonia is separated from the other gases.

(2)

- (b) What happens to the unreacted hydrogen and nitrogen?

(1)

The equation for the reaction is:



The forward reaction is exothermic.

- (c) Calculate the volume of ammonia produced from the complete reaction of 825 dm³ of hydrogen.

Volume of ammonia = _____ dm³

(2)

- (d) The Haber process uses a temperature of 450 °C and a pressure of 200 atmospheres.

Why are these conditions used?

Tick **two** boxes.

A higher pressure is maintained using less energy

☐

A higher temperature would increase the equilibrium yield

☐

A lower pressure would decrease the equilibrium yield

☐

A lower temperature would make the reaction too slow

☐

There are more product molecules than reactant molecules

☐

(2)

Most of the ammonia produced is used to make fertilisers.

Table 2 shows information about compounds used as fertilisers.

Table 2

Compound	Formula	Cost in £ / tonne
A	NH_4NO_3	220
B	$(\text{NH}_4)_2\text{HPO}_4$	350
C	KCl	235

- (e) Which element in compound A improves agricultural productivity?

(1)

- (f) Which **two** compounds can be mixed to make a fertiliser containing three elements that improve agricultural productivity?

Give a reason why you have chosen these compounds.

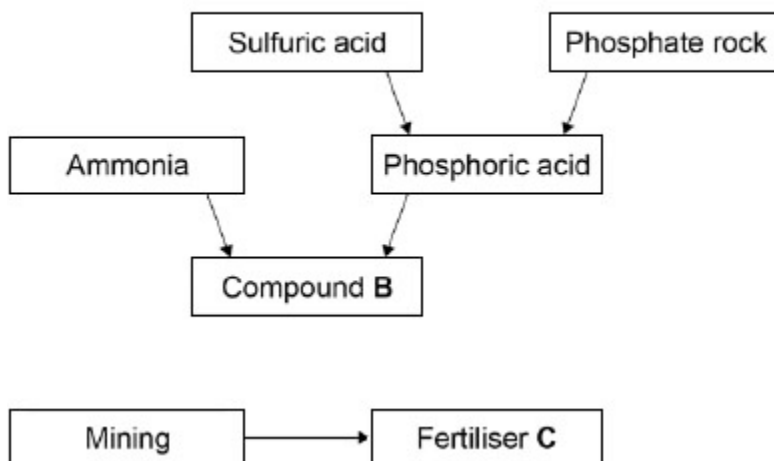
Compounds _____ and _____

Reason _____

(2)

- (g) **Figure 2** shows a flow chart for the production of compounds B and C.

Figure 2



Suggest **two** possible reasons for the difference in cost between compounds **B** and **C**.

1. _____

2. _____

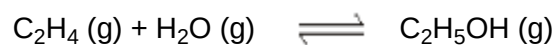
(2)

(Total 12 marks)

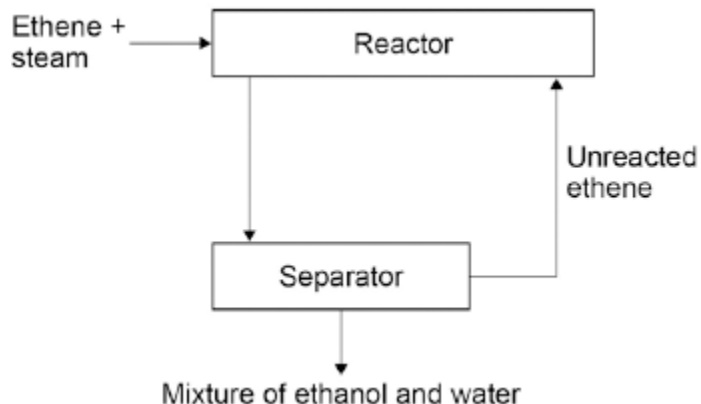
4.

In industry ethanol is produced by the reaction of ethene and steam at 300°C and 60 atmospheres pressure using a catalyst.

The equation for the reaction is:



The figure below shows a flow diagram of the process.



- (a) Why does the mixture from the separator contain ethanol and water?

(1)

- (b) The forward reaction is exothermic.

Use Le Chatelier's Principle to predict the effect of increasing temperature on the amount of ethanol produced at equilibrium.

Give a reason for your prediction.

(2)

- (c) Explain how increasing the pressure of the reactants will affect the amount of ethanol produced at equilibrium.

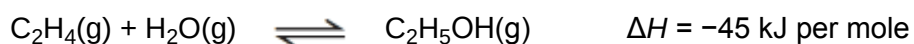
(2)

(Total 5 marks)

5.

A company manufactures ethanol ($\text{C}_2\text{H}_5\text{OH}$).

The reaction for the process is:



The temperature and pressure can be changed to increase the yield of ethanol at equilibrium.

- (a) Explain what is meant by equilibrium.

(3)

- (b) (i) How would increasing the temperature change the **yield** of ethanol at equilibrium?

Give a reason for your answer.

(2)

(ii) How would increasing the pressure change the **yield** of ethanol at equilibrium?

Give a reason for your answer.

(2)

(c) A catalyst is added to increase the rate of the reaction.

Explain how adding a catalyst increases the rate of a chemical reaction.

(2)

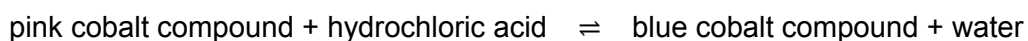
(Total 9 marks)

6.

Cobalt forms coloured compounds.

A pink cobalt compound reacts with hydrochloric acid.

The reaction can be represented as:



The forward reaction is endothermic.

When both cobalt compounds are present in a solution at equilibrium, the equilibrium mixture is purple.

(a) What is meant by equilibrium?

(2)

- (b) The equilibrium mixture is cooled.

Explain what happens to the concentration of the pink cobalt compound.

(3)

- (c) More hydrochloric acid is added.

Explain what happens to the colour of the equilibrium mixture

(3)

- (d) Why does cobalt form different coloured compounds?

(1)

- (e) An oxide of cobalt has the formula Co_2O_3

Which cobalt ion is present in this oxide?

Tick (✓) **one** box.

Co^+ ☐

Co^{2+} ☐

Co^{3+} ☐

Co^{4+} ☐

(1)

- (f) Cobalt compounds can act as catalysts.

Which two statements about cobalt compounds are correct?

Tick (✓) **two** boxes.

They allow reactions to reach equilibrium more quickly.

☐

They are reactants in reactions catalysed by cobalt compounds.

☐

They are used up when acting as catalysts.

☐

They increase the equilibrium yield of reactions.

☐

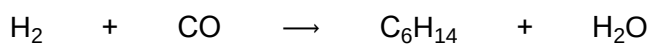
They provide a different reaction pathway.

☐

(2)

- (g) The reaction of hydrogen with carbon monoxide is catalysed by cobalt metal.

Balance the equation for the reaction.



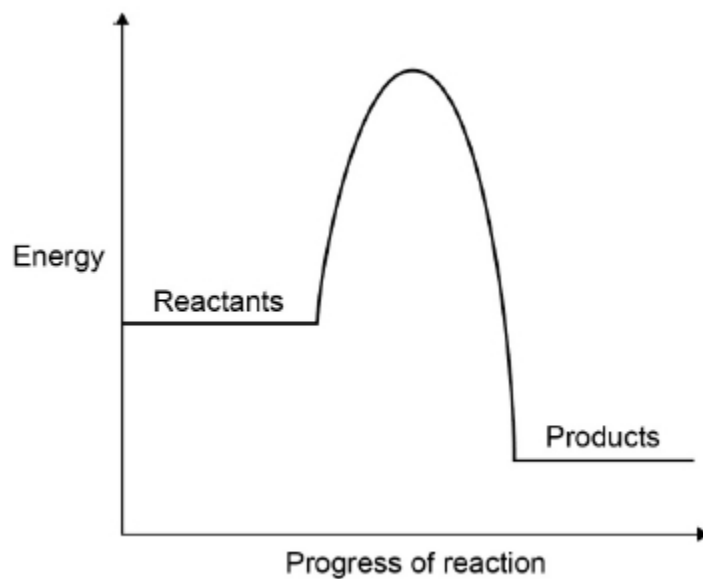
(1)

(h) C_6H_{14} is an alkane.

What is the formula of an alkane containing 18 hydrogen atoms?

(1)

(i) The graph shows a reaction profile diagram for a reaction **without** a catalyst.



On the graph:

- draw the reaction profile diagram for a catalysed reaction
- draw and label an arrow to show the activation energy for the reaction **without** a catalyst.

(2)

(Total 16 marks)

Mark schemes

1.

(a) water

allow H₂O

1

(b) becomes (more) red

1

(because the position of) equilibrium moves to the right

allow (because) the concentration of FeSCN²⁺ (ions) increases

allow (because) the forward reaction is favoured

1

(so that) the (increase in the) concentration of thiocyanate (ions) is reduced

allow (so that) the increase in the concentration of thiocyanate (ions) is counteracted

1

(c) (the position of) equilibrium moves to the left

allow the concentration of Fe³⁺ (ions) increases

allow the reverse reaction is favoured

1

(so that) the (increase in the) temperature is reduced

allow (so that) the increase in the temperature is counteracted

1

(therefore) the forward reaction is exothermic

allow (therefore) the forward reaction releases energy (to the surroundings)

1

(d) no change in equilibrium position

1

(because) no gases are present

allow (because) only aqueous solutions are present

1

(e) Co²⁺

1

[10]

2.

(a)

an answer of 17.6470588 (%) correctly rounded to at least 2 significant figures scores 2 marks

$$\frac{6}{34} \times 100$$

1

$$= 17.6 (\%)$$

allow 17.6470588 (%) correctly rounded to at least 2 significant figures

1

(b)

*allow converse arguments in terms of higher pressure
ignore references to rate*

higher yield (of hydrogen or carbon monoxide or product)

allow more hydrogen or more carbon monoxide or more product

allow equilibrium moves to the right

allow equilibrium moves in the forward direction

1

(because) fewer moles / molecules / particles on left hand side

or

(because) more moles / molecules / particles on right hand side

allow (because) the reverse reaction produces fewer moles / molecules / particles

or

allow (because) the forward reaction produces more moles / molecules / particles

*do **not** accept fewer / more atoms*

1

(c) no effect (on yield of hydrogen)

allow position of equilibrium unaffected by pressure

ignore references to rate of reaction

1

(d)

an answer of 2.25 scores 3 marks

350 (°C) and 285 (atmospheres) = 63 (%)

and

450 (°C) and 200 (atmospheres) = 28 (%)

allow a value between 62 (%) and 64 (%) inclusive

1

$\frac{63}{28}$

allow a correct expression using incorrectly determined value(s) for percentage yield

1

= 2.25 (times greater)

allow a correct calculation using incorrectly determined value(s) for percentage yield correctly evaluated and rounded to at least 2 significant figures

1

(e)

allow converse arguments in terms of low(er) pressure

any **one** from:

- the energy costs would be high(er)
ignore energy / cost unqualified
- the equipment would need to be strong(er)
allow the equipment would be (more) expensive (to build / maintain)
- high(er) pressures are (more) dangerous
allow (more) dangerous because (greater) risk of explosion

1

(f) higher temperatures produce a lower (percentage) yield (of ammonia)

allow converse

allow correct reference to shift in equilibrium

ignore references to pressure

1

(g) world population has increased

1

any **one** from:

- demand for fertiliser has increased
allow more food needed
- increased demand for other specified ammonia-based products e.g. nitric acid, drugs, dyes, explosives

1

[12]

- 3.** (a) cool 1
- to $-34\text{ }^{\circ}\text{C}$
- allow temperatures below $-34\text{ }^{\circ}\text{C}$ but above $-196\text{ }^{\circ}\text{C}$* 1
- (b) recycled (to the reactor) 1
- (c) $825 \times \frac{2}{3}$ 1
- $= 550\text{ (dm}^3\text{)}$ 1
- an answer of 550 (dm³) scores 2 marks*
- (d) a lower pressure would decrease the equilibrium yield 1
- a lower temperature would make the reaction too slow 1
- (e) nitrogen / N 1
- (f) **B** and **C** 1
- contain nitrogen, phosphorus and potassium 1
- (g) **(B)**
- any **two** from:
- more stages
 - uses more energy
 - uses more raw materials
 - takes longer
- allow converse for C* 2
- [12]
- 4.** (a) both water vapour and ethanol will condense
- allow steam for water vapour*
- allow they both become liquids*
- allow ethane condenses at a lower temperature*
- allow some of the steam hasn't reacted*
- allow it is a reversible reaction / equilibrium* 1
- (b) amount will decrease 1

because the equilibrium will move to the left

1

- (c) more ethanol will be produced

1

because system moves to least / fewer molecules

1

[5]

5.

- (a) the forward and backward reactions occur

allow reversible

1

at (exactly) the same rate

1

in a closed system

allow therefore the concentrations / amounts of the reactants and products remain the same

1

- (b) (i) increasing the temperature would lower the yield of ethanol **or** the (position of) equilibrium moves to the left

if student has stated that increasing the temperature increases the yield then award 0 marks

1

since the backwards reaction is endothermic **or** the forward reaction is exothermic

1

- (ii) increasing the pressure would increase the yield of ethanol **or** the (position of) equilibrium moves to the right

if student has stated that increasing the pressure decreases the yield then award 0 marks

1

because the position (of equilibrium) moves in the direction of the lower number of moles (of gas)

2 (moles / molecules / volumes / particles) on lhs / 1 (mole / molecule / volume / particle) on rhs

1

- (c) (a catalyst) provides an alternative pathway

1

with lower activation energy

or

(a catalyst) lowers the activation energy (1)

so less energy is needed to react **or** more particles react (1)

1

[9]

6.

(a) in a closed system

1

the rate of the forward and backward reactions are equal

1

(b) concentration increases

1

(because) reaction / equilibrium moves to the left / reactant side

1

(since the) reverse reaction is exothermic

allow (so that) temperature increases

1

(c) becomes blue

1

(because) reaction / equilibrium moves to the right / product side

1

(so) concentration of blue cobalt compound increases

allow (so that) concentration of hydrochloric acid decreases

1

(d) (cobalt has) ions with different charges

allow (cobalt is a) transition metal

1

(e) Co^{3+}

1

(f) they allow reactions to reach equilibrium more quickly

1

they provide a different reaction pathway

1

(g) $13\text{H}_2 + 6\text{CO} \rightarrow \text{C}_6\text{H}_{14} + 6\text{H}_2\text{O}$

allow multiples

1

(h) C_8H_{18}

1

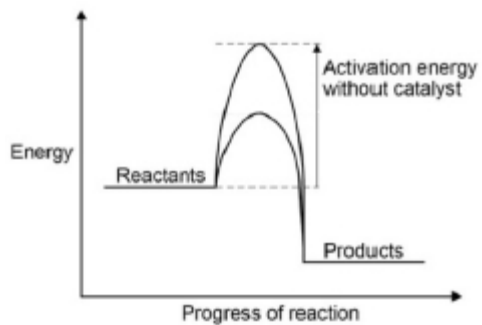
- (i) curve below printed curve
*do **not** accept different reactant or product levels*

1

vertical arrow from reactant level to peak of **printed** curve

1

an answer of:



scores 2 marks

[16]

Examiner reports

1.

- (a) A minority of students identified water as the solvent.
- (b) The mark for describing the colour change was scored more often than the mark for the shift in the position of equilibrium that caused this colour change. The mark for the application of Le Chatelier's Principle was rarely scored by students.
- (c) Once again, the shift in equilibrium position and the identification of the forward reaction as exothermic were seldom justified in terms of Le Chatelier's Principle. The distribution of marks awarded to students was very similar to that for (b).
- (d) Students answered this question poorly. A minority recognised that the position of equilibrium had not changed, but fewer still linked the non-effect of pressure with the absence of gases. A common error was to state that there there are equal numbers of moles on each side of the equation, as though this **were** a gaseous reaction.
- (e) A minority of students were able to identify cobalt as the only transition metal ion in the list.

2.

- (a) Nearly half of the students scored both marks. Students who provided incorrect responses either ignored the stoichiometry of the equation or did not know the expression which represents atom economy.
- (b) The idea that lower pressure increases the yield was understood by many of the students but some had difficulty explaining why. Around a third of the students scored both marks.
- (c) Around a third of the students scored this mark. The fact that there was the same number of moles on each side of the equation was often ignored by students. Instead, these responses focused on an increase in pressure increasing the rate of reaction and therefore assumed that this would increase the yield.
- (d) This was well done by many of the students although reading the graph was problematic for some. Misinterpreting the y-axis scale and/or using a wrong curve were the most frequently seen sources of error. Around two-thirds of the students scored all three marks.
- (e) Most students recognised that 285 atmospheres is more expensive but many failed to explain why in terms of energy or cost of plant construction. There were also many vague references to danger or safety. Around a third of the students scored this mark.
- (f) Around half of the students scored this mark, having interpreted the graphs correctly and recognised that increasing / decreasing the temperature decreases / increases the yield of ammonia.
- (g) Many students recognised that the world population had increased since 1950 and that the demand for food had also increased. Students who did not gain the second mark did not relate the increase in demand for ammonia to an ammonia-based product. Around 40% of the students gained both marks.

5.

- (a) Only a small proportion of students scored all three marks, which was disappointing as this question was straight recall from the specification. Students had a good knowledge of the forward rate being equal to the backward rate, although the idea of a closed system was seen rarely. Often students used vague statements such as 'the forward reaction is the same as the reverse reaction' and as such only gained a single mark. Many students thought that the idea of an equilibrium must mean that there were equal quantities of matter on each side of the equation.
- (b) (i) This question was answered in a variety of ways, with a large number of students indicating that the yield would decrease, although their explanation did not always link to equilibrium. Many stated that the yield would increase and based that explanation on the temperature increasing the rate of reaction. The better responses included a correct explanation of the change in equilibrium, as well as the correct effect on the rate of the reaction. About half of students scored both marks.
- (ii) The number of students who understood the role of the number of particles on each side of the equation was very encouraging, with about half of students scoring both marks in this demanding question. Many students gave a sound explanation to describe the increase in yield in terms of Le Chatelier's Principle. However, many responses recognised the increase in yield but then proceeded to explain why an increase in pressure would increase the rate of the reaction. Many students in both (i) and (ii) did not appreciate that there is a difference between yield and rate. The better students gave a response that included a correct explanation of the change in equilibrium and an explanation of the correct effect on the rate of the reaction.
- (c) About half of students gained both marks, usually by stating that there was an alternative pathway of lower activation energy. Some students thought that a catalyst gave particles more energy and the language at times was very confused, describing particles having energy above that of the activation energy.